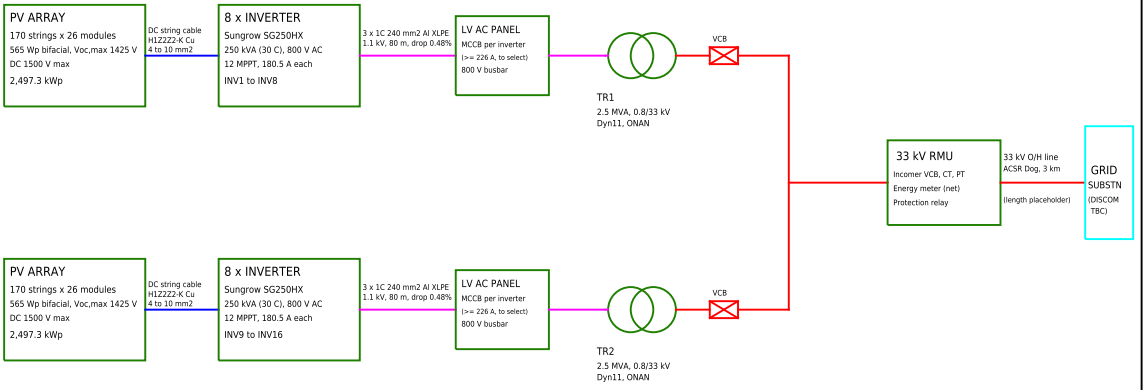
5 MWp SOLAR PV PLANT: SINGLE-LINE DIAGRAM (PRELIMINARY)

Two identical 2 MW blocks (INV1-8 + TR1, INV9-16 + TR2) join at the 33 kV RMU. Ratings are design values, not vendor-confirmed.



NOTES

1. DC/AC ratio 1.249; 340 strings (148 MPPT inputs with 2 strings, 44 with 1) across 16 inverters.
2. Net energy at 33 kV point 6,917 MWh/yr after 2.28% AC-side losses (cables, transformers, 33 kV line).
3. Not yet shown: auxiliary supply, SCADA/communication, weather station, earthing and lightning protection, surge protection devices, DC isolation detail. Add before any construction use.
4. MCCB/relay ratings and cable lengths are to be confirmed with the final layout and vendor data.

LEGEND

- DC (1500 V max)
- LV AC (800 V)
- 33 kV

PROJECT	5 MWp ground-mount solar PV plant, West Bengal (resource site: Kolkata)
DESCRIPTION	Single-line diagram
DRAWING NO.	SPV5-DWG-02-SLD-R0
SCALE	Not to scale
DRAWN / DATE	Sagnik Kumar Nath / 28-10-2026
STATUS	PRELIMINARY, NOT FOR CONSTRUCTION